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Received: 2 February 2026 | Accepted: 8 September 2026

DOI: 10.1111/2041-210x.70419

Methods in Ecology and Evolution

APPLICATION

HGTScanner: Detecting horizontal gene transfer at the fine scale using whole organelle genomes

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Handling Editor: Itay Mayrose

Abstract

1. Horizontal gene transfer (HGT) has profoundly shaped eukaryotic organellar genomes like plant mitochondria. Yet no generalized tools are currently available to efficiently characterize HGT across these genomes due to both high structural complexity and low sequence divergence compared to prokaryotic HGTs. Accurate identification of HGT, including the genomic location and donor lineage, is critical for understanding its impact on genetic novelty and genome evolution as well as species interactions that generate HGTs.
2. Here, we introduce HGTScanner, a Python tool to characterize HGT across whole organellar genomes. HGTScanner automatically establishes homology, produces alignments and phylogenies, and tests hypotheses of HGT based on BLAST, phylogeny and synteny. It contains a built-in database of green plant (Viridiplantae) organellar sequences to facilitate the characterization of promiscuous mitochondrial HGTs and mitochondrial plastid DNAs (MTPTs) in plants but may also be adapted to other organisms like fungi which share many similar genomic features.
3. Using benchmark data from the early diverging angiosperm *Amborella* and parasitic plant Orobanchaceae, we demonstrate that HGTScanner can effectively locate and identify the donor lineages of MTPT and HGT. In *Amborella*, we recovered the well-documented fusion of moss and green algae mitochondria, and discovered prevalent HGTs from the New Caledonian parasitic plant *Hachettea* (Mystropetalaceae). In the parasitic Orobanchaceae, non-native MTPTs are strongly linked with their sympatrically distributed host plants.
4. HGTScanner is an efficient and flexible tool that can drive discoveries of HGT without a priori knowledge regarding the donor identity. The sympatric distribution of donor-recipient pairs in HGT events suggests that HGTs can serve as valuable genetic markers for organismal interactions across time and terrain.

KEYWORDS
Amborella, mitochondria, non-coding region, parasite, phylogeny, plastid, synteny

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1 | INTRODUCTION

Horizontal gene transfer (HGT) occurs when one organism transfers genetic information to another—potentially distantly related—organism that is not its offspring. HGT has long been recognized as a major driver of genome evolution in prokaryotes and to an increasing degree in eukaryotes (Graham & Davies, 2021; Keeling, 2024; Kurland et al., 2003; Richardson & Palmer, 2007). In plants, HGT has been found in nuclear, plastid and mitochondrial genomes (mitogenome) of ferns, gymnosperms and angiosperms (Janouškovec et al., 2013; Li et al., 2014; Palmer et al., 2000; Rice et al., 2013; Richardson & Palmer, 2007; Won & Renner, 2003). In animals, classic examples include the transfer of lectin-like antifreeze proteins between fish (Graham et al., 2008) and the transfer of carotenoid biosynthesis genes from fungi to pea aphids (Moran & Jarvik, 2010). The growing availability of eukaryote genomes represents an opportunity to systematically quantify HGT abundance and precisely characterize their genome locations, functions and donor lineages. Knowledge about the identity and origin of HGTs provides insight into cellular mechanisms that drive genetic innovation, the molecular basis of novel functions and natural histories of species interaction (e.g. Cai et al., 2021; Hepburn et al., 2012; Li et al., 2014).

Plant mitochondria represent a particularly valuable system to tackle these questions and explore methods for three reasons. First, plant mitochondria are known to contain a rich inventory of horizontally acquired DNA and frequently involve ecological interactants (Richardson & Palmer, 2007). Other types of organellar genomes, including plant plastids and animal mitochondria, exhibit profound stasis where HGT is extremely rare (Keeling & Palmer, 2008). For example, the New Caledonian angiosperm *Amborella* acquired entire mitochondrial genomes from likely epiphytic green algae and moss (Bergthorsson et al., 2004; Rice et al., 2013; Taylor et al., 2015), and parasitic plants frequently harbour host-derived mitochondrial sequences (Petersen et al., 2020; Richardson & Palmer, 2007; Sanchez-Puerta et al., 2017; Xi et al., 2013). Characterizing HGT donor lineages can provide valuable data points for the taxonomy, timing and location of organismal interactions across evolutionary time (Cusimano & Renner, 2019). Second, the typically sub-megabase genome makes mitochondrial genomics more accessible than full nuclear genomes, requiring modest computational resources and sequencing investment while allowing for sampling tens of thousands of taxa. Finally, plant mitochondria are structurally comparable and evolutionarily homologous to many other organismal systems exhibiting similar HGT features. Specifically, fungal mitochondria and prokaryotic genomes frequently contain HGTs from diverse donors that contribute to the reshuffling of their genomes (Koonin et al., 2001; Kouvelis et al., 2023; Rosewich & Kistler, 2000). So methods developed for plant mitochondrial HGT are potentially applicable to fungal, animal and even prokaryotic systems. Furthermore, an estimated 90% of plant mitochondrial genomes are made up of intergenic regions (Mower et al., 2012), presenting an interesting technical challenge and allowing for the exploration of effective methods for homology establishment across thousands of genomes with rampant structural rearrangements.

Current methods for HGT detection vary widely in terms of scope, approach and rigour (see recent reviews in Dutton & Reiter, 2023; Mariault et al., 2025). Higher genetic similarity to distant rather than closely related taxa is a widely recognized indicator for HGT (Richardson & Palmer, 2007). This can be established by sequence similarity-based metrics such as the Alien Index calculated from BLAST scores (Boschetti et al., 2012; Gladyshev et al., 2008), genomic synteny (Adato et al., 2015) or through discordant phylogenetic signals (Yuan et al., 2023). Sequence similarity-based metrics are particularly popular for detecting HGTs in bacteria and archaea because habitual cross-kingdom transfers create high sequence divergence that is readily identifiable by BLAST. These BLAST-based methods may be supplemented with GC bias and motif analyses to further aid in HGT detection (e.g. Bertelli & Brinkman, 2018; Garcia-Vallve et al., 2003; Vernikos & Parkhill, 2006). However, BLAST-based HGT identification methods rely on the assumption that substantial sequence differences will have accumulated between the transferred gene and its native copy. Applying such methods to slowly evolving sequences, such as plant mitochondria or even HGT events from fungal and animal mitochondrial DNA, can introduce substantial challenges (Koutsovoulos et al., 2022). Without an explicit evolutionary framework, it is also impossible to determine the direction of HGT or pinpoint the exact donor and recipient of historical HGTs (e.g. HGT from the common ancestor of clade A to clade B). In addition, most current HGT identification tools in prokaryotes are optimized toward protein BLASTp scores (e.g. Li et al., 2018; Nguyen et al., 2015; Podell & Gaasterland, 2007; Zhu et al., 2014), which restricts their application to coding regions only. No current tools are designed to handle the rampant genome restructuring in plant mitochondrial genomes.

An explicitly phylogenetic-based approach represents a more rigorous HGT hypothesis testing framework by examining the placement of the query sequence in relation to its taxonomy. Such approaches have been employed since early attempts to identify HGT in plants, including the explosive invasion of a group I intron in the mitochondrial *cox1* gene (Cho et al., 1998). Incorporating phylogeny offers additional flexibility in the type of genetic loci included and the taxonomic scale of the analyses. Along these lines, many tools have been developed to use gene tree species tree reconciliation to identify HGT (e.g. Boc et al., 2012; David & Alm, 2011; Morel et al., 2024; Yuan et al., 2023). However, the requirement of homology establishment and a reference species tree once again has largely restricted standalone reconciliation methods to coding regions. In particular, none of the existing tools are particularly useful for HGT characterization in plant mitogenomes due to their complex genome evolution (Palmer et al., 2000; Smith & Keeling, 2015). Plant mitogenomes contain a rich inventory of DNA that has been exchanged within a cell, within an organism or between organisms. Genomic synteny can be rapidly lost within a few million years among closely related species (Sloan et al., 2012). Therefore,

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even for adjacent genomic loci, the genealogy background can vary substantially in terms of their cellular compartment source (e.g. plastid versus mitochondrion) and the taxa from which they originate (Figure 1). The combination of these phenomena can generate phylogenetic topologies that are difficult to interpret. Moreover, in exploratory studies where the donor lineage is unknown or highly divergent, standard reconciliation methods require either a comprehensively sampled universal species tree encompassing hundreds of thousands of potential donors or manual construction of reference trees for each locus. For modern high-throughput genomic analyses, none of these options is realistic.

Here, we present HGTScanner, an empirically motivated Python tool for efficient and effective identification of HGTs, particularly in non-coding regions of organellar genomes. It is freely available on GitHub under the GNU Licence (https://github.com/lmcai/HGTScanner). It includes a built-in database that facilitates characterization of plant mitochondrial HGTs from unknown donors or cellular compartments, but HGTScanner can also take custom reference databases to investigate HGTs from other organellar genomes and taxon groups. It automatically establishes homology across the whole genome, scaffolds rearrangements, performs alignment and phylogeny inference and finally evaluates HGT based on exclusive BLAST hits, phylogeny and local synteny (see definition of synteny metrics below) in relation to user-supplied taxonomy. It is computationally efficient and can be implemented on laptops. The software is written in Python 3 and its dependencies include pybedtools, pandas, numpy, statistics, hmmlearn, ete3, BLAST+, MAFFT and FastTree (Camacho et al., 2009; Dale et al., 2011; Harris et al., 2020; Huerta-Cepas et al., 2016; Katoh & Standley, 2013; Price et al., 2010). It has been tested on macOS and Linux and is compatible with the Windows Subsystem for Linux (WSL).

2 | METHODS DESCRIPTION

2.1 | General recommendations for experimental design

HGTscanner identifies HGT by exclusive sequence homology or high phylogenetic affinity between distantly related taxa, with additional evidence from local synteny. Therefore, the confidence of HGT detection increases with the phylogenetic distance between putative donor and recipient lineages. In general, cross-order, cross-kingdom and inter-organellar (e.g. mitochondrial plastid DNA (MTPT)) transfers introduce highly distinct sequences relative to the recipient. These cases are among the easiest to identify both in terms of the sequences and lineages involved. In contrast, gene transfers between closely related species, such as those within the same family, require more comprehensive taxon sampling to mitigate inaccurate HGT identification. In such cases, the discordance between a gene tree and species tree used as evidence for HGT may also originate from gene loss, incomplete lineage sorting, introgression or analytical processes like phylogenetic estimation error.

HGTScanner is designed to characterize transfers among distantly related lineages at the family level. Although the definition of family as a taxonomic unit is subjective, we found it useful and appropriate in practice. Below this level, incomplete taxon representation of mitochondrial sequences in GenBank makes it difficult to identify donor lineages at the genus level with confidence. Above the family, order-level identification can span tens of thousands of globally distributed taxa, making it uninformative for the study of organismal interactions using HGT at fine temporal and spatial scales. For more complex cases where donors and recipients are sister families or even sister genera (Gawryluk et al., 2019), great caution is

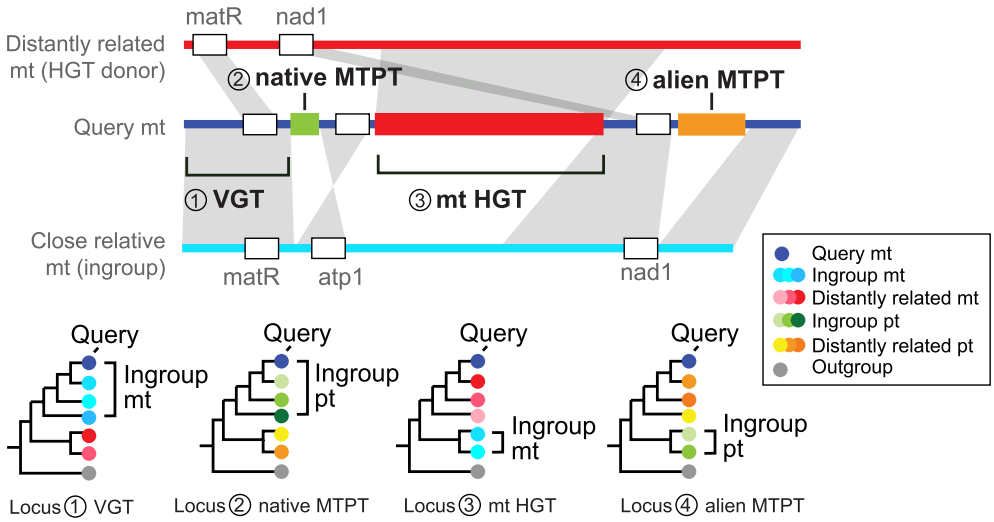


FIGURE 1 The complex structure and genetic backgrounds of plant mitochondrial sequences with expected phylogeny pattern of each locus type. Plant mitochondrial genomes contain generally conserved gene regions and highly variable intergenic regions that consist of mitochondrial plastid DNAs (MTPTs), repeats, and horizontally transferred sequences (HGTs). Vertically inherited regions (VGT) are expected to show higher synteny with closely related ingroup mitochondrial (mt) sequences, while the insertion of plastid (pt) sequences and HGT can break the synteny. VGT, MTPT and HGT also generate distinct expectations on the placement of the query sequence on a phylogeny, which can be used for hypothesis testing.

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needed to make claims regarding HGT. Additional evidence from large-scale genome synteny and statistical comparisons of HGT frequency with control species may be useful, but the output from HGTScanner alone is insufficient.

Carefully selected taxon sampling is the most important step in HGT evaluation. HGTScanner measures the similarity of the query sequence to potential donors versus to its close relatives, the latter of which are referred to as 'ingroup' (Gladyshev et al., 2008). As a result, the selection of user-defined ingroup taxa is critical for mitigating false positives in HGT identification. For HGTScanner, these ingroups do not need to be the sister clade of the query taxa or form a clade, but they should show high genetic similarity to the query taxa in terms of homologue presence and sequence identity. For example, Nymphaeaceae may serve as an ingroup for Amborellaceae even though all extant angiosperms are equally distantly related to Amborellaceae (Chase et al., 2016). HGTScanner takes a conservative approach by assigning any query sequences showing higher affinity to the ingroup on a phylogeny, either as sister or paraphyletic close relatives, as vertically inherited instead of HGT. Thus, the broader the ingroup is defined taxonomically, the more stringent the criteria can be for HGT identification. Yet this also comes with an important limitation and caveat of HGTScanner: the criteria for 'vertical inheritance' are quite generous in HGTScanner and may contain true HGTs from especially ingroup taxa.

2.2 | The built-in Viridiplantae database

To facilitate HGT detection in plant mitochondria, we assembled a built-in Viridiplantae database that will be regularly updated. This includes all 14,707 NCBI RefSeq complete plastid genomes and 68,519 plant mitochondrion sequences from a total of 24,661 species in NCBI (accessed Nov 6, 2025). The mitochondrial database was filtered from 85,889 raw records of NCBI plant mitochondrial sequences to remove sequences <100 bp, with ambiguous taxonomic identification (e.g. no genus-level identification), or with identical duplicate sequences from the same species. Lists of species for both datasets are provided in Tables S1 and S2 and on GitHub, enabling users to assess whether additional sequences should be generated—either by adding more ingroup species to stabilize the phylogeny or by incorporating expected donor lineages to more finely resolve the origins of HGTs.

2.3 | How HGTScanner works

For each query sequence, vertical inheritance generates the expectation that homologous regions of this sequence are present among ingroups and exhibit phylogenetic affinity consistent with the species tree. Violation of these expectations, including absence of homology in the ingroup or phylogenetic clustering of the query sequence with distantly related lineages, can serve as evidence of HGT in HGTScanner.

HGTScanner first establishes homology between the query sequence and tens of thousands of highly heterogeneous plant mitogenomes from the built-in database. To accomplish this, an initial BLASTN search with the Discontiguous MegaBLAST algorithm (–task dc-megablast) is performed to identify homologous sequences and syntenic blocks. We tested several whole genome alignment tools including Mauve and Cactus (Darling et al., 2004; Paten et al., 2011), but found that they were computationally too demanding and often neglected small-scale syntenic blocks (e.g. 100–150 bp) where HGTs often occur. We thus chose to use BLASTN results directly. Users may supply a bed file of mitochondrial exons or repeats to mask these regions in HGTScanner, which is optional in most species but helps break up large-sized mitochondrial genomes like *Amborella* (see case study below). The masked query sequence is then BLAST against the entire Viridiplantae mitochondrial sequence database from NCBI GenBank using an e-value threshold of $1e^{-20}$. The lower BLAST e-value threshold is applied because it can provide strong evidence for HGT when exclusive homology is found in distantly related lineages but not ingroups, which is especially important for utilizing the non-coding regions of mitochondria. The user can supply additional sequences (–mt_add_seq) from ingroup taxa or expected HGT donor lineages that are not present in the built-in database.

To establish boundaries of genetic loci for subsequent phylogenetic assessment while taking into account the rampant genome rearrangements in the mitochondria, BLAST hits are ordered based on their genomic location and merged into longer synteny blocks using the pybedtools Python module by requiring at least 50 bp overlap (bedtools merge -d – 50) (Dale et al., 2011). Fragmented regions were scaffolded to align with the orientation of the query sequence (Figure 2). In the case where the orthologous copy is lost, this scaffold step may generate chimeras with non-orthologous sequences; however, we found that non-orthologous chimeras impacted less than 1% of loci and can be sufficiently mitigated by sampling more diverse ingroups. Each initial synteny block is further divided if the following criteria are simultaneously met: (1) contains >85% missing data; (2) contains fewer than 5 long hits spanning >60% of the total region; (3) contains BLAST hits from more than 100 species; (4) spans >600 bp. These criteria are effective in identifying large alignments with obvious recombination boundaries, but leave small blocks with few BLAST hits as-is. Locus subdivision is also essential for the identification of shorter HGTs nested within long synteny blocks. To divide long synteny blocks, we applied the Hidden Markov Models implemented in the hmmlearn Python module to identify recombination boundaries repeatedly seen in BLAST hits (Figure 2). By applying the above scaffolding steps, the query mitochondrial genome is typically divided into hundreds to thousands of loci, each of which is homologous to at least one additional species. Sequences from each synteny block were subsequently aligned using MAFFT using the query as the reference (--6merpair --keeplength --addfragments), and a maximum likelihood phylogeny was inferred by the approximately maximum-likelihood method implemented in FastTree v2.2 (Price et al., 2010). The phylogeny is then rooted based on the family level relationship from the NCBI Common Taxonomy Tree (https://

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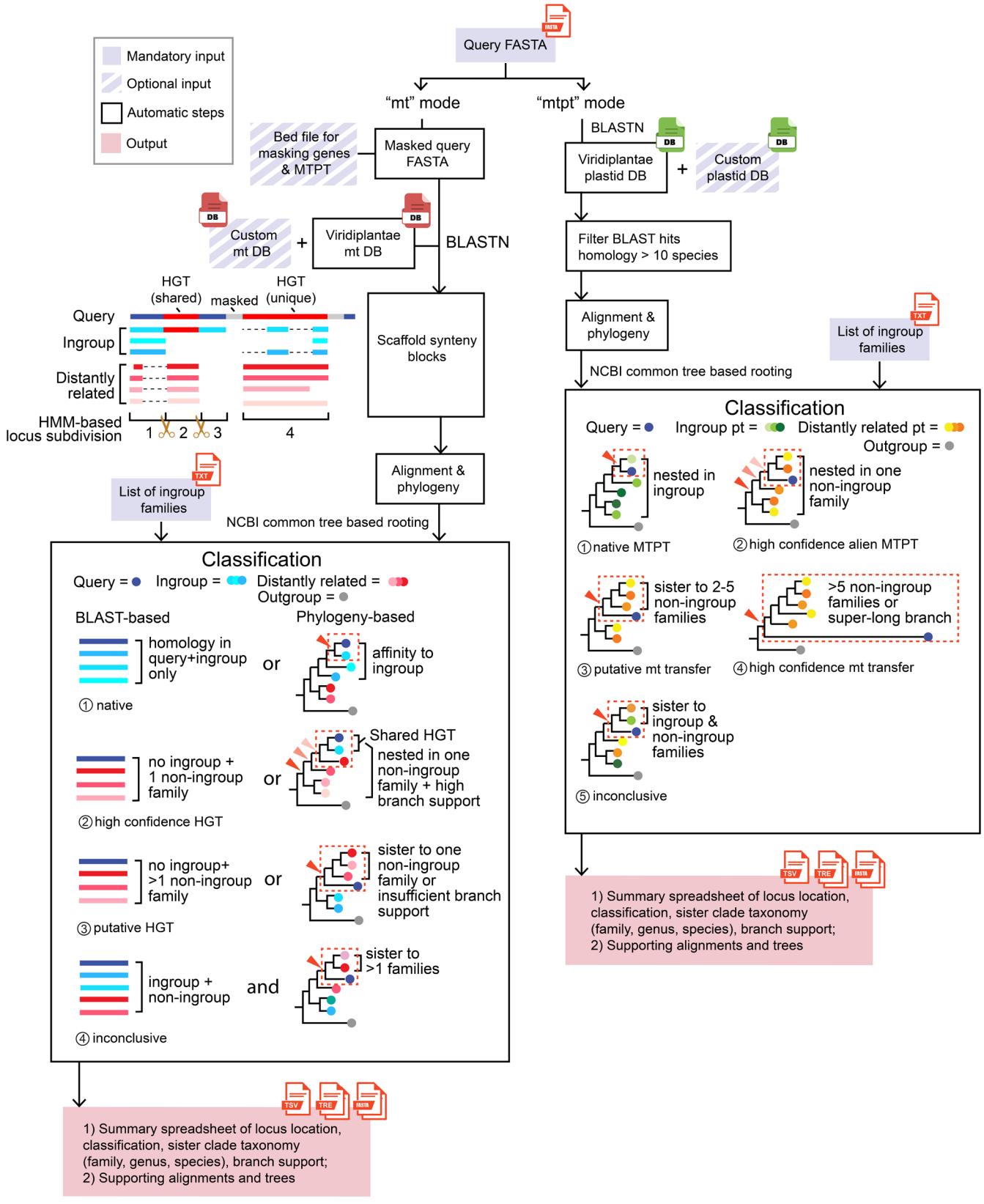


FIGURE 2 Overview of the HGTScanner pipeline for automated identification of MTPT and HGT events.

www.ncbi.nlm.nih.gov/Taxonomy/CommonTree/wwwcmt.cgi) and the branch support is evaluated by the Shimodaira–Hasegawa test in FastTree (Shimodaira & Hasegawa, 1999). Users can also use IQ-TREE or RAxML (Stamatakis, 2014; Minh et al., 2020) instead for phylogenetic inference using the '-notree' flag and distribute the tasks to high-performance computer clusters. We recommend

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a two-stage workflow for most studies: using the default FastTree analyses to perform initial screening, followed by genome-wide or focal-region-only re-analysis using IQ-TREE or RAxML for more accurate topological inference and branch support prior to biological interpretation.

Mitochondrial HGT is subsequently classified based on BLAST and phylogenetic evidence with additional support from synteny (Figure 2). Specifically, within a rooted gene tree, HGTScanner will visit the successive sister and parent branches of the query sequence iteratively from its immediate sister to more distantly related lineages. The taxonomic affiliations of these successive sisters are compared to the list of ingroup species provided by the user to determine whether HGT can be supported. In addition, this iterative tree browsing also allows for the detection of ancient HGTs shared by multiple species when they cluster with the query branch and nest within a larger donor clade. This technology has been used by several phylogeny-explicit HGT detection tools such as AvP and HGTphyloDetect to utilize the phylogenetic information without requiring a reference species tree for full-scale reconciliation (Becq et al., 2010; Koutsovoulos et al., 2022; Yuan et al., 2023). The following criteria are then used for classifying the query into four categories: 'native', 'high-confidence HGT', 'putative HGT' and 'inconclusive' (Figure 2). It is important to note that these classifications are rule-based categories rather than statistically calibrated confidence levels. Users are encouraged to adjust or apply additional filtering criteria, such as the phylogenetic support values and synteny metrics, using the information in the output spreadsheet.

I. Native: (1) Homology is only found in the query and ingroup families (BLAST-based evidence); (2) the query is phylogenetically affiliated with ingroup as an immediate sister (forming a clade) or as an adjacent successive sister (not forming a clade, relaxed phylogeny-based evidence). Users can apply additional branch support cutoff for phylogeny-based identification using the output spreadsheet.
II. High-confidence HGT: (1) Homology is only found in the query family and one other non-ingroup family (BLAST-based evidence); (2) the query is nested within a non-ingroup family with >0.95 branch support in FastTree (stringent phylogeny-based evidence). We also recommend setting the branch support cutoff at 85 for RAxML and 95 for IQ-TREE via the -bs flag. These branch support cutoffs roughly control the false positive rate of phylogenetic estimation under 5% for each programme (Anisimova et al., 2011; Minh et al., 2013).
III. Putative HGT: (1) Homology is not found in any other ingroup families but in multiple non-ingroup families (BLAST-based evidence). Phylogeny is then applied to identify the potential donor lineage. (2) The query is nested within a non-ingroup family but branch support is <0.95 in FastTree (phylogeny-based evidence). (3) The query is placed as sister to a single non-ingroup family, but not fully nested within (phylogeny-based evidence). Users can apply additional branch support cutoff for phylogeny-based identification using the output spreadsheet.
IV. Inconclusive: Homology is present in both ingroup and outgroup families. The sister group of the query contains more than one non-ingroup family and thus cannot assign the donor of HGT confidently.

The final output contained a summary spreadsheet (*.hgt.sum.tsv) of the locus coordinates, classification, sister taxa (family, genus, species), support value, and additional synteny information. Synteny metrics include the median coverage of the query locus found in putative donor versus ingroup lineages, and the number of fragments that these syntenic regions consist of. A continuous, high-coverage syntenic region in the distantly related donor lineage, contrasted with a fragmented and gappy alignment in the ingroup, provides additional support for HGT inference. The output folder (*_HGTscanner_supporting_files) contained sequence alignments and phylogenies for each locus. Sequence names consisted of three fields: family, scientific name, original sequence ID and location (i.e. Family|Genus_species|seqID_start1_end1+start2_end2+...). The location information included in the sequence header can be used to calculate custom syntenic metrics by placing different levels of penalty for missing data, fragmentation and inversion.

Characterization of mitochondrial plastid DNA (MTPT) follows a highly similar process and can be invoked by the '-m mtpt' flag. The major difference is that (i) we applied a representative non-redundant plastid database containing 3722 genera from 518 families for MTPT identification; (ii) the homologue scaffolding step is omitted because plastid genome synteny is highly conserved; and (iii) high sequence divergence and phylogenetic uncertainty of the query sequence suggest plastid-to-mitochondrion-to-mitochondrion transfer instead of direct plastid-to-mitochondrion transfer and are thus classified as putative or high-confidence 'ancestral mitochondrial transfer' (Figure 2). For a detailed description of MTPT identification workflow and classification criteria, see Note S1 in the supplementary material.

3 | APPLICATION EXAMPLES

3.1 | MTPT identification in parasitic Orobanchaceae

A pilot version of HGTScanner was applied to a dataset of 43 Orobanchaceae species to reveal the RNA-mediated mechanism of MTPT (Cai et al., 2026). Without repeating these results, here we demonstrate the use of HGTScanner by characterizing MTPT in four Orobanchaceae species. *Mannagettaea hummelii* and *Aeginetia indica* are non-photosynthetic parasites specialized on Fabaceae and Poaceae, respectively. We thus expect alien MTPT from the corresponding host and an overall lower abundance of MTPT due to their reduced plastid genome sizes and transcriptional activities (Cai et al., 2026). *Alectra sessiliflora* and *Rehmannia glutinosa* are photosynthetic members of Orobanchaceae and we expect more native MTPT and an overall higher abundance of MTPT.

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The input files included a FASTA file of the mitochondrial assembly from each species and a tab-delimited file of closely related families where we listed all Lamiales families as ingroup (taxon.txt). To run HGTScanner, the following command was used:

```
$ python HGTscanner.py -m mtpt -q Mannagettaea_
hummelii.fas -o Mhu -taxon taxon.txt -pt_add_seq
Orobanchaceae_pt.fas
```

HGTScanner identified 14 to 26 MTPT-like loci in each species. Phylogenetic assessments sampled 200 genera and 47 families per locus on average (Data S1). The results revealed that non-photosynthetic holoparasites generally had fewer MTPTs per species but more alien MTPTs from host plastids compared to photosynthetic species (Table S3; Data S1), which was consistent with the trend we reported in our empirical study (Cai et al., 2026). For *Mannagettaea hummelii*, HGTScanner identified 5 native MTPTs, 2 as high-confidence alien MTPTs well-nested within Fabaceae, 6 ancestral mitochondrial transfers, and 1 inconclusive (Table S3). Specifically, the alien MTPT pointed to *Caragana* (Fabaceae) as the donor (Table S3), which was a documented host and grew sympatrically with *Mannagettaea hummelii* in Southern Siberia and Northwest China (Novopokrovskii & Tzvelev, 1958; Zhang & Tzvelev, 1998).

For the Poaceae specialist *Aeginetia indica*, HGTScanner identified 2 high-confidence alien MTPTs from Poaceae, 4 native MTPTs, 8 ancestral mitochondrial transfers and 2 inconclusive (Table S3). In hemiparasite *Alectra sessiliflora*, HGTScanner identified 3 high-confidence alien MTPTs from Euphorbiaceae and Malvaceae, 15 native MTPTs, 5 ancestral mitochondrial transfers and 3 inconclusive cases. For the free-living *Rehmannia glutinosa*, HGTScanner identified 9 native MTPTs, 11 ancestral mitochondrial transfers, and 3 inconclusive cases (Table S3). No high-confidence alien MTPT was identified in the free-living species *Rehmannia glutinosa*.

3.2 | Benchmarking MTPT identification

Accuracy of classification—We evaluated the quality of HGTScanner results by searching the candidate MTPT loci in the NCBI nr/nt database (accessed Dec 11, 2025) and manually examining the taxonomy and genomic location of top hits. These BLAST hits were effective in distinguishing true MTPT versus mitochondrion-vectored transfers because the latter were expected to have top hits from mitochondria instead of plastids (Mishra & Lercher, 2025). After excluding all inconclusive results, 5 of the 79 loci (93.2%) had concordant classifications between HGTScanner and manual verification, with most (4 out of 5) misclassifications falling under the 'native MTPT' category (Table S3). Manual NCBI verification showed that all 4 cases of misclassified 'native MTPT' involved mitochondrion-to-mitochondrion transfer of native plastid sequences instead of direct plastid-to-mitochondrion transfers. These events would have been challenging to distinguish without comparative mitochondrial sequences, which are currently not included under the 'mtpt' function. The fifth case of misclassification (Locus #36 of *Alectra sessiliflora*) originated from the inaccurate tree topology inferred by FastTree: The query sequence was placed as sister to a single species of Setchellanthaceae (*Setchellanthus caeruleus*) and then they nested within a clade of 40 Malvaceae species (Data S1). More accurate topology inferred by IQ-TREE correctly placed the query sequence within the Malvaceae clade without Setchellanthaceae and this locus was classified as 'high-confidence alien MTPT' (Table S3). Finally, 6 loci were designated as 'inconclusive' due to uncertain phylogenetic placements. Manual verification demonstrated that all 6 cases were results of ancestral mitochondrial transfers, where the query sequences were more similar to mitochondrial sequences of close relatives or alien donors rather than any plastid.

3.2.1 | Improved classification quality with IQ-TREE

Phylogenetic application using IQ-TREE improved the proportion of concordant unambiguous predictions from 93.2% to 98.8% (Table S3), likely due to more accurate estimation of topology and branch lengths compared to FastTree (Price et al., 2010; Zhou et al., 2018). Specifically, IQ-TREE inferred longer branches for ancestral mitochondrial transfers, which allowed HGTScanner to distinguish them from true 'native MTPTs' and corrected 3 out of 4 misclassifications compared to the FastTree result. In addition, the accurate IQ-TREE topology for Locus #36 of *Alectra sessiliflora* also allowed HGTScanner to correctly assign this locus as 'alien MTPT' from Malvaceae, as discussed above.

Computational time—When running MTPT analysis on a 2023 MacBook Pro with Apple M3 Pro chip, each plant mitochondrial genome on average took about 2–14 min of wall-clock time and less than 400 Mb peak RAM to complete with a single thread (Table 1). This included BLASTN against the database, alignment, FastTree phylogeny and classification. Specifically, it took 121.84–335.24 s CPU time and 145.3–381.2 Mb peak RAM for the four species we tested, and between 112.88 and 934.68 s for all 43 Orobanchaceae included in our empirical study (Figure 3a). The computational resources scaled linearly with the size of the input genome (Figure 3a) and were strongly influenced by the number of loci subject to alignment and phylogenetic inference (Figure 3b). The peak RAM was also positively correlated with input genome size but was below 300 Mb for most species (Figure 3c). Phylogenetic inference with IQ-TREE required up to 40 times more CPU time on the same device compared to the default FastTree (Table 1). For the four species we tested, it took 3243.74–14,058.23 s CPU time (54 to 123 min wall-clock time) and 151.1–206.0 Mb peak RAM on a laptop. We thus strongly recommend that users distribute IQ-TREE or RAxML jobs to high-performance computing clusters.

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TABLE 1 Computational time and peak RAM of MTPT identification in HGTScanner when using FastTree or IQ-TREE for phylogeny inference.

Experiment	Species	Query genome size (bp)	Number of MTPT loci	Wall-clock time (s)	CPU time (s)	Peak RAM (mb)
MTPT-FastTree	*Mannagettaea hummelii*	539,045	14	123.85	121.54	381.2
MTPT-IQTREE	*Mannagettaea hummelii*	539,045	14	3539.05	4518.91	151.1
MTPT-FastTree	*Aeginetia indica*	420,362	16	121.84	112.88	145.3
MTPT-IQTREE	*Aeginetia indica*	420,362	16	3243.74	3687.95	178.3
MTPT-FastTree	*Alectra sessiliflora*	469,124	26	335.24	329.97	205.6
MTPT-IQTREE	*Alectra sessiliflora*	469,124	26	14,058.23	13,349.94	206.0
MTPT-FastTree	*Rehmannia glutinosa*	547,032	23	224.13	214.27	176.1
MTPT-IQTREE	*Rehmannia glutinosa*	547,032	23	7400.91	7256.17	178.4

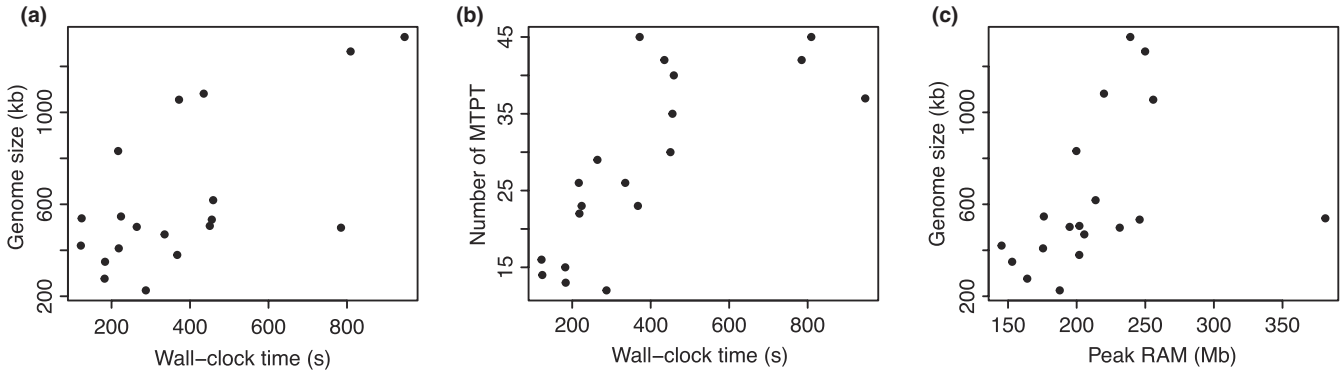


FIGURE 3 Computational time and peak RAM of MTPT identification in HGTScanner scale positively and linearly with the query genome size. The wall-clock time and peak RAM were recorded by analysing the empirical data from (Cai et al., 2026) on a 2023 MacBook Pro with Apple M3 Pro chip. (a) Positive correlation between input genome size and wall-clock computational time. (b) Positive correlation between the number of MTPTs and wall-clock computational time. (c) Positive correlation between input genome size and peak RAM usage.

3.3 | Example workflow to identify HGT in the *Amborella* mitochondrion

Mitochondrial HGT in *Amborella trichopoda* has been carefully researched (Bergthorsson et al., 2004; Rice et al., 2013; Taylor et al., 2015) and can be used to both demonstrate the utility of HGTScanner and evaluate its performance. To characterize mitochondrial HGT in *Amborella*, we designated all families in the ANA-grade and magnoliids as ingroup (Chase et al., 2016), where they might form clades or successive paraphyletic groups to *Amborella* in empirical gene trees. In preliminary tests, restricting the ingroup to the ANA-grade alone resulted in considerable HGTs from Lauraceae and Piperales (magnoliids). Although Rice et al. (2013) acknowledged potential HGTs from magnoliids, we adopted a conservative approach to assign these early diverging angiosperms as ingroup. This approach may result in false negatives for true magnoliid HGTs but reduces spurious inferences. We first ran HGTScanner under the 'mtpt' mode to identify any true MTPTs to be masked:

```
$ python HGTscanner.py -m mtpt -q Amborella_
trichopoda_mito.fasta -o AtrMTPT -taxon taxon.txt
```

This step took a longer 1534.46 CPU time to complete due to the exceptional 3.9 Mb mitochondrial genome of *Amborella*. HGTScanner identified 46 native MTPTs with an average length of 1557 bp (Figure 4a; Table S4), which is significantly longer than the 885 bp MTPTs found in Orobanchaceae. This aligned with the general recognition that the lower rate of mitochondrial genome rearrangements in *Amborella* preserved longer synteny for alien DNAs (Rice et al., 2013). In addition, we also found two adjacent 668-bp-block high-confidence alien MTPT well-nested within a clade of 5 Elaeocarpaceae genera (Figure 4a; Table S4). Around 15 loci showed strong phylogenetic affiliation with various algae clades (e.g. Locus #10, 37, 67; Table S4). Most of these loci were classified as 'ancestral mitochondrial transfer' because the query sequence had long branch length and was placed sister to multiple algae families instead of nested within any of them.

We generated a bed file to mask genes and MTPTs (Atr.mask.bed) and used the following command to detect HGT:

```
$ python HGTscanner.py -m mt -q Amborella_
trichopoda_mito.fas -o Atr -taxon taxon.txt -b
Atr.mask.bed -bs 0.95
```

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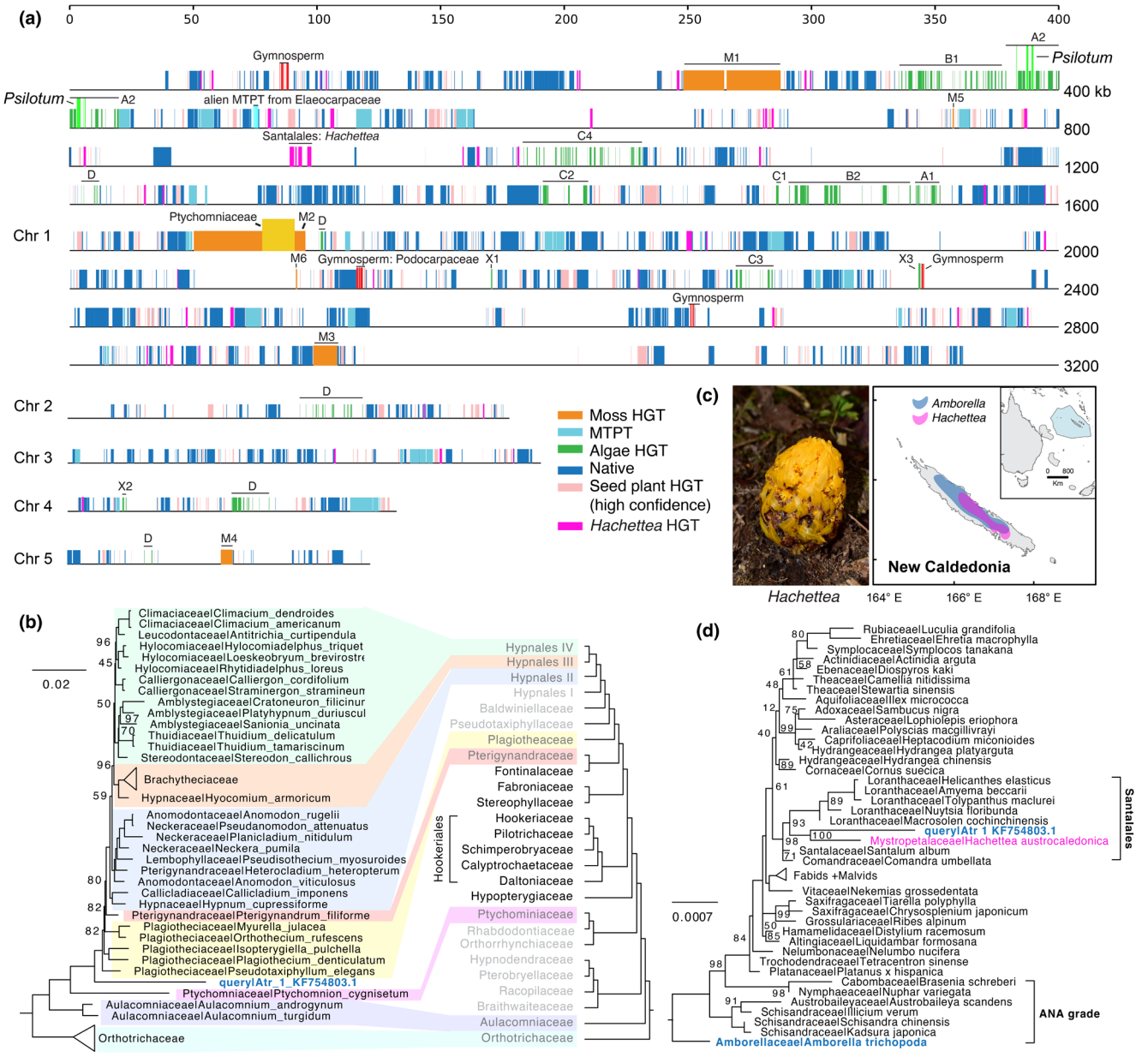


FIGURE 4 Foreign DNA in the *Amborella* mitochondrial genome identified by HGTScanner. (a) Chromosome maps of the *Amborella* mitochondrion are plotted in a similar style to Rice et al. (2013) to facilitate comparison. Coloured regions highlight native DNA (navy blue), mitochondrial plastid DNA (MTPT; skyblue), moss HGT (orange), algae HGT (green) and high-confidence seed plant HGTs identified by HGTScanner (salmon). The moss and algae genomes are labelled the same as Rice et al. (2013) (A-D, and M genomes), but we identified 5 additional small-scale HGTs not characterized before (X1-X3 for algae and M5-M6 for moss). For seed plant HGTs, we found prevalent DNA from a New Caledonian root parasite *Hachettea* (Mystropetalaceae, Santalales), which likely corresponds to the Santalales HGT identified in Rice et al. (2013). We also highlight 4 long gymnosperm HGT blocks spanning multiple loci. (b) Maximum likelihood phylogeny of potential Hookeriales HGTs inferred from concatenated sequences of M1 genome. Branch support was measured by 1000 ultrafast bootstrap in IQ-TREE and only values below 100 were shown. Phylogenies were pruned to show representative lineages only. For full size tree, see Data S1. The moss phylogeny on the right was modified from Li et al. (2024) to show the possible donor of HGT from unsampled lineages including Hookeriales. (c) Overlapping range of *Hachettea* and *Amborella* in New Caledonia. *Hachettea austrocaledonica* image credit: Joey Santore (CC BY-NC 4.0). (d) Maximum likelihood phylogeny of a *cob*-like locus showing HGT from *Hachettea* to *Amborella*. Branch support was measured by 1000 ultrafast bootstrap in IQ-TREE and only values below 100 were shown. Phylogenies were pruned to show representative lineages only. For full size tree, see Data S1.

This step took 5782.16 s CPU time (5988.08 s wall-clock time) and 1.19 Gb peak RAM. HGTScanner annotated a significantly higher proportion of the mitogenome (42.0%) compared to the previous study (30.1% in figure S4 of Rice et al., 2013). For the 373 annotated HGTs from Rice et al. (2013) (GenBank accession KF754799-KF754803), HGTScanner recovered 67 of the 68 moss HGTs (blocks M1-M4 in Rice et al., 2013) and 160 of the 163 algal HGTs (blocks A-D). The missed HGTs were all tRNAs too short to generate significant BLAST e-values

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in HGTScanner but were nested within the corresponding long HGT blocks. Moreover, HGTScanner identified 5 previously missed small-scale HGTs, including 2 moss HGTs (M5 and M6) and 3 algal HGTs (X1-X3) (Figure 4; Table S5). Due to the great expansion of moss and algal mitochondrial genome resources over the past decade, we were able to refine the HGT boundary, synteny and donor lineage (Figure 4; Table S5). For the 142 angiosperm HGTs, however, HGTScanner classified 52 of them as vertically inherited. This discrepancy largely reflected our expanded definition of ingroup because 75.0% of these loci were associated with magnoliids, which were recognized by Rice et al. (2013) as plausible HGT donors and were not in this study.

All four major moss HGTs M1-M4 showed near-perfect synteny with the moss donor, forming 39, 45, 9 and 4 kb tracks (Figure 4a; Table S5). On average, each moss HGT retained 94.3% of the homologous region from the donor genome, typically represented by 1.7 fragments, whereas only 30.0% of the corresponding region was detectable in ingroups and they were highly fragmented into an average of 8.3 segments (see synteny metrics in Table S5). The boundaries of these HGTs aligned well with Rice et al. (2013). The placements of the *Amborella* query sequence were consistent in M1 and M3-M6 (Figure 4c), where it formed successive sister clades with Hypnales and Ptychomniaceae in (Ptychomniaceae, (*Amborella*, Hypnales)). Because *Amborella* was never sister to or nested within a single family in these loci, they were classified as 'Putative HGT' or 'Inconclusive' (Table S5). On the other hand, a 31.2 kb region within M2 (Figure 4a; locus #1248 in Data S1) confidently placed *Amborella* as sister to Ptychomniaceae (IQ-TREE support 100). These results were consistent with the gene-based study from Taylor et al. (2015), where both Ptychomniales and Hookeriales contributed to the *Amborella* moss HGTs. Specifically, the dominant (Ptychomniaceae (*Amborella*, Hypnales)) placement aligned with the anticipated position of Hookeriales (Li et al., 2024) even though mitogenome data from this moss clade are currently lacking (Figure 4b). Many of these candidate donors, including Ptychomniaceae and Hookeriales, have Gondwanan origin and are widely distributed in New Caledonia, New Zealand and Australia (Ho, 2010; Huttunen et al., 2012), which partially overlapped the range of *Amborella*.

Unlike moss, algae HGTs formed long but fragmented syntenic blocks (Figure 4a). Rice et al. (2013) attributed these algal HGTs to four potential algal donors named A–D, based on their gene order, GC content and intron composition. The two major algae HGT tracks identified by HGTScanner—the 83 kb B1 + A2 and the 66 kb C1 + B2 + A1 (Figure 4a)—aligned well with the boundaries of Rice et al. (2013) and consisted of 59 and 28 fragmented HGT blocks, respectively (Table S5). The unmapped regions were largely intergenic spaces that did not show homology to any known NCBI nt/nr core sequences (accessed Dec 25, 2025). Within B, C, D genomes, the majority (72.3%) of individual 'high-confidence HGTs' were from Watanabeaceae, whereas only 22.1% HGTs found in the A genome were Watanabeaceae (Table S5). Thus, if B, C, D algal HGTs came from three different species (Rice et al., 2013), they were likely all members of Watanabeaceae, which includes lichen photobionts and widespread freshwater algae.

The HGT tracks from the A genome, on the other hand, were particularly heterogeneous. Despite a small proportion of Watanabeaceae HGTs (e.g. the 1410 bp locus #342 in Table S5), donors of most HGT loci were identified to be a mix of many distantly related taxa including Chlorodendraceae, Pleurastraceae, Oocystaceae, Trebouxiaceae, Coccomyxaceae and Botryococcaceae (Table S5). The collective diversity of donor lineages in the A genome contradicted with the hypothesis from Rice et al. (2013) that it represented a single and likely unsampled donor deeply embedded within the algal phylogeny, although this possibility could not be fully ruled out. Photobiont algae from Trebouxiaceae and Coccomyxaceae are known to live as mixed colonies in the same lichen thallus (Dědková et al., 2025). Thus, pre-HGT recombination in the donor organism or a third-party vector might be a more plausible explanation. For the A genome, it might either contain many HGTs from various algae taxa prior to the transfer, or the integration of the A genome might be vectored by a third media such as (cyano)bacteria, plasmid or mycobionts where recombination took place. Along these lines, the algal A2 region was admixed with DNAs from the whisk fern (*Psilotum*, Psilotaceae) where 7 loci totalling 4541 bp showed high phylogenetic affinity between *Amborella* and *Psilotum* (Figure 4a; locus #329, 332–335, 344–346 in Table S5). Here, *Amborella* and *Psilotum* typically formed a clade sister to the common ancestor of Coccomyxaceae and Botryococcaceae (Data S1). *Psilotum* thus may have acted as a vector facilitating algal HGT to *Amborella*. Alternatively, this could reflect repeated algae-to-vascular-plant HGTs from similar algal donors to *Psilotum* and *Amborella*. An alternative to the pre-HGT mixing hypothesis is the post-HGT recombination model proposed by Taylor et al. (2015), where homologous recombination brought distantly-located HGTs into continuous tracks in the *Amborella* mitogenome.

Finally, we analysed seed plant derived HGTs in *Amborella*. In contrast to the continuous angiosperm HGT tracks proposed by Rice et al. (2013), these HGTs were generally short (<1000 bp), scrambled and derived from diverse source species (Table S5). Many HGTs in gene-like regions overlapped with previous reports from Bergthorsson et al. (2004) and Rice et al. (2013). For example, locus #1 was a nad1-derived sequence transferred from Betulaceae as reported by Bergthorsson et al. (2004). We also found previously unrecognized donor lineages including 4 gymnosperm HGTs (e.g. Podocarpaceae) spanning multiple continuous loci in 1.8–3.7 kb tracks (Figure 4a; Table S5). Among all seed plants, the family and genus contributing the highest number of HGT loci were the parasitic Mystropetalaceae (Santalales) and its member *Hachettea* (Figure 4a,b), which likely corresponded to the broadly defined Santalales HGTs from Rice et al. (2013). We identified 62 HGT loci totalling 24.1 kb from *Hachettea*, the majority of which ($n = 48$) were supported by exclusive homology. The longest *Hachettea* HGT block spanned 8 continuous loci in an 8.4 kb track (locus #663–670 in Table S5). Phylogenetic analyses of gene regions like *cob* (locus #431 in Table S5), on the other hand, further supported the placement of the donor as *Hachettea* within Santalales (Figure 4d).

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The fine placement of *Hachettea* as a dominant HGT donor for *Amborella* has several important implications. First, the significant DNA exchange between *Amborella* and *Hachettea* potentially suggests a parasite–host relationship between the pair, which has not yet been recorded in the field. In fact, little is known regarding the host preference of *Hachettea*, but its New Zealand relative *Dactylanthus* (Mystropetalaceae) has been found to grow on Araliaceae and Podocarpaceae (Parker, 2015). We advocate for future herbarium collections and botanical studies of parasitic plants to include host observations. Second, the restricted, endemic and sympatric distribution range of *Amborella* and *Hachettea* in New Caledonia further supported their ecological interaction, either directly as host–parasite or vectored by a third party. This highlights the potential of HGTs to serve as calibration points for divergence time and biogeographic inference, especially for parasitic plants lacking fossil records. Relative dating based on HGT has been successfully utilized to elucidate the time tree of microbes and fungi (Davín et al., 2018; Szánthó et al., 2025; Wolfe & Fournier, 2018), and we anticipate its application in plant evolution (Cai, 2023). Third, although HGTScanner was designed for family level identification of HGT donors, the example of *Hachettea* suggested that dense sampling or complementary evidence from natural history may refine donor identity to the genus or even species level (*Hachettea* comprises a single species *Hachettea austrocaledonica*). Nevertheless, even with exhaustive sampling of extant taxa, the possibility of extinction should always be considered especially for historical events. Finally, phylogenies from locus #431 (cob-like, Figure 4d), #1422 (rpl16-like) and #2289 (ndh1-like) all supported *Hachettea*-to-*Amborella* transfers, while #1549 (sdh3-like) was a more conventional *Amborella*-to-*Hachettea* transfer (Data S1). If they indeed represent rare parasite-to-host HGTs, as have been occasionally reported (Davis et al., 2005; Mower et al., 2004; Rice et al., 2013), the directionality of HGTs involving host–parasite pairs may be more fluid than we think before.

4 | LIMITATIONS

HGTScanner is optimized to detect HGT at the family level. Given the rate of sequence divergence in plant mitochondria and other eukaryotic organellar genomes, this taxonomic resolution prioritizes minimizing false positives, but it also limits the types and ages of HGT events that can be confidently identified. For example, 'high-confidence HGT' is assigned only when the query is phylogenetically nested within a single donor family, which biases this category toward recent events. Ancient HGT events involving donors ancestral to multiple families or extinct/unsampled lineages are often ambiguously classified by HGTScanner (e.g. Hookariales HGT in *Amborella*). The age constraints also apply to the recipient: historical HGT shared by a few recipient genera can be correctly recognized, but those taking place prior to the origin of the family are typically reported as native. On the other hand, when the recipient belongs to an early diverging monotypic family such as Amborellaceae, the absence of closely related ingroups can result in unstable phylogenetic placement and lack of homology at native loci, which may lead to false positive HGT detection. Thus, an ideal scenario for HGTScanner involves a recent transfer between members of multi-genus families, in which both native and horizontally acquired copies are well-nested within their respective families. For users interested in older transfers or events involving extinct or unsampled donors, the phylogenies, summary tables and prior taxonomic knowledge of the donor (e.g. order- or class-level affinities) can be leveraged to guide hypothesis-driven analyses. Along these lines, we strongly encourage users to customize classification criteria and interpret the HGT predictions in the context of genome structure and macrosynteny, life-history strategy and biogeographic patterns.

Phylogenetic and BLAST-based donor assignment also provides limited resolution below the genus level (Adato et al., 2015). For instance, we could not determine whether the B, C, and D algal genomes originated from the same Watanabeaceae species or not. In specific cases, the combination of dense taxon sampling, sufficient sequence divergence, repeated transfers and overlapping geographic distribution may allow genus-level donor inference (e.g. *Hachettea* HGT in *Amborella*). However, fine-scale donor identification generally requires additional lines of evidence, such as genomic synteny or GC content, as demonstrated by Rice et al. (2013).

Finally, we anticipate that the Viridiplantae database integrated within HGTScanner will be regularly updated as GenBank sequence collection expands. At present, the plastid database is relatively well-represented for vascular plants, but sampling of bryophytes and algae remains sparse. The mitochondrial database remains largely incomplete, capturing only 2.8% of species-level diversity within Viridiplantae (One Thousand Plant Transcriptomes Initiative, 2019). These limitations should be considered when interpreting results. Nevertheless, even with incomplete reference databases, HGTScanner provides a useful first-pass framework for donor lineage identification, as illustrated by the moss-derived HGTs in *Amborella*. Targeted follow-up sampling can then be used to generate custom reference sequences and rerun HGTScanner for refined donor identification.

5 | CONCLUSIONS

While there is an increasing interest and effort to understand the evolution of plant mitogenomes, their highly variable architecture and expansive intergenic regions continue to pose major challenges for assembly, annotation and comparative analyses. Specifically, the intergenic sequences have long been regarded as dispensable elements that host alien genetic materials from other cellular compartments and species. Our new tool HGTScanner enables researchers to examine the sources of these non-native DNAs with curated built-in databases. The default analysis using FastTree can be completed on a standard laptop within 10 min to one hour of wall-clock time for most plant mitochondrial genomes smaller than 1 Mb. Runtime can be substantially longer and may require access to high-performance computing clusters for analyses using IQ-TREE and on larger genomes. By substantially lowering computational and technical

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barriers, HGTScanner facilitates the characterization of MTPTs and HGT when reporting new plant mitochondrial genomes, especially in species prone to incorporate HGT. As we have demonstrated with the example from *Amborella* and parasitic plants, these non-native DNAs embody histories of species interactions in historical time. Many donors involve the known ecological partners of these species, including epiphytes and parasitic plant hosts. We propose that these HGTs may serve as a new type of genetic marker for the study of symbiotic interactions.

AUTHOR CONTRIBUTIONS

Conceptualization: Liming Cai. Data curation: Liming Cai. Methodology: Liming Cai. Software: Liming Cai. Visualization: Liming Cai. Case studies: Liming Cai. Writing—original draft: Liming Cai and Rachel O Cohen. All authors contributed critically to the drafts and gave final approval for publication.

ACKNOWLEDGEMENTS

The author would like to thank members of the Cai Lab for valuable comments during the development of the tool.

FUNDING INFORMATION

The study was supported by the startup fund from the University of Florida and Stengl-Wyer postdoctoral fellowship from the University of Texas at Austin to Liming Cai. The computational resources supporting this work were provided by the HiperGator Research Computing at the University of Florida.

CONFLICT OF INTEREST STATEMENT

None declared.

PEER REVIEW

The peer review history for this article is available at https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/2041-210x.70419.

DATA AVAILABILITY STATEMENT

HGTScanner is an open-source Python program (GNU Licence) accessible on GitHub (https://github.com/lmcai/HGTScanner). Software documentation can be found on the GitHub repository. The published version of HGTScanner is archived on Zenodo [https://doi.org/10.5281/zenodo.22033400] (Cai & Cohen, 2026a). No new sequence data are generated from this study. All alignments and phylogenies supporting the findings are deposited on Zenodo [https://doi.org/10.5281/zenodo.22033346] (Cai & Cohen, 2026b).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Data S1. Input and output files of HGTScanner analyses performed in the current study. This included mitochondrial plastid DNA (MTPT) characterization in *Mannagettaea hummelii*, *Aeginetia indica*, *Alectra sessiliflora*, and *Rehmannia glutinosa*; MTPT and HGT characterization in *Amborella trichopoda*; and assessment of moss and algae donors of *Amborella* HGTs using concatenated sequences of A-D and M genomes in IQ-TREE. Data can be accessed via Zenodo (https://doi.org/10.5281/zenodo.22033346).

Note S1. Workflow of mitochondrial plastid DNA (MTPT) identification in HGTScanner.

Table S1. Taxonomy and GenBank accession number of Viridiplantae plastid genomes from GenBank RefSeq included in the built-in database of HGTScanner. Accessions included in the 3722-genus representative database are noted in the column 'In_default_rep_db'.

Table S2. Taxonomy and GenBank accession number of Viridiplantae mitochondrial sequences from GenBank included in the built-in database of HGTScanner.

Table S3. Results and accuracy assessment of mitochondrial plastid DNA (MTPT) classification by HGTScanner when using FastTree or IQ-TREE. Accuracy is evaluated by manual inspection of NCBI nr/nt BLAST results and phylogeny.

Table S4. Mitochondrial plastid DNA (MTPT) classification results of *Amborella trichopoda* mitochondrial genome by HGTScanner. For supporting alignments and phylogenies, see Data S1.

Table S5. Mitochondrial HGT identification summary output of *Amborella trichopoda* by HGTScanner. Moss and algae HGTs aligning with Rice et al. (2013) are annotated in the Note column. For supporting alignments and phylogenies, see Data S1.

How to cite this article: Cai, L., & Cohen, R. O. (2026). HGTScanner: Detecting horizontal gene transfer at the fine scale using whole organelle genomes. *Methods in Ecology and Evolution*, *00*, 1–14. https://doi.org/10.1111/2041-210x.70419

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